Arista Access Node Recovery Procedure for Hardware/Environmental Faults (ALARM_5375)

Node: BAS-ACC-62 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_5375

Description

The occurrence of hardware or environmental faults, such as ASIC memory parity errors, line card crashes, or power supply redundancy loss on an Arista access node, specifically BAS-ACC-62 situated at the Basavanagudi POP, necessitates a comprehensive understanding of the underlying protocol and hardware mechanics. The Arista EOS (Extensible Operating System) powering these devices is designed to provide high availability and redundancy. However, when such faults occur, they can significantly impact network stability and performance. ASIC memory parity errors, for instance, can lead to unpredictable behavior, including packet drops or corruption, while line card crashes can cause direct service interruptions. Power supply redundancy loss, although less immediately disruptive due to the presence of redundant power supplies, poses a significant risk if not addressed promptly, as it reduces the overall resilience of the system. The SOP outlined here is designed to methodically diagnose and resolve these issues, leveraging Arista's CLI to execute precise diagnostic and resolution commands, thereby ensuring the swift restoration of full operational capacity to the affected access node.

Pre-requisite Checks

- Dispatch remote hands to Basavanagudi POP for on-site inspection and intervention.

- Verify the status of redundant links to ensure no single points of failure exist.

- Check environmental conditions (temperature, humidity) to ensure they are within the specified operating range for the equipment.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Errors	High
Line Card Hardware Failure	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

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Command: show hardware forwarding ascii, Expected Output: Detailed forwarding table information, including any errors or inconsistencies., Description: This command provides insight into the data plane's forwarding behavior, helping isolate issues related to ASIC memory parity errors or line card malfunctions.
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```
Command: show boot variables, Expected Output: Boot configuration and variables, useful for identifying potential issues in the boot process., Description: Examines the boot process for any anomalies that could indicate hardware or firmware issues.

Command: show environment, Expected Output: Environmental readings, including temperature and power supply status., Description: Checks the environmental conditions and power supply status to identify any issues that could be contributing to the fault.
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Resolution Steps

Step 1: Step: Reset Line Card, Command: hardware module reset, Description: Resets the line card, which can resolve issues caused by transient hardware faults or software glitches.

Step 2: Step: Replace Faulty Power Supply, Command: no power supply redundancy, Description: Temporarily disables power supply redundancy to allow for the safe replacement of a faulty power supply unit, followed by re-enabling redundancy once the replacement is complete.

Step 3: Step: Update Firmware, Command: swi update, Description: Updates the firmware to the latest version, which may include fixes for known issues related to ASIC memory parity errors or other hardware faults.

Post-Check Verification

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Command: show interfaces counters, Expected Output: Interface statistics, including error counters., Description: Verifies that interfaces are operating correctly and that error counters are not incrementing after the resolution steps.

Command: show boot variables, Expected Output: Updated boot configuration reflecting any changes made during the resolution process., Description: Confirms that any necessary changes to the boot process have been successfully applied.

Command: show environment, Expected Output: Environmental readings indicating healthy operation., Description: Ensures that environmental conditions are within acceptable ranges and that power supplies are operating redundantly.
```

Roll-back Procedures

- Step: Revert Firmware, Command: swi rollback, Description: Rolls back the firmware to the previous version if the update caused unforeseen issues, restoring the device to its pre-update state.

- Step: Restore Original Configuration, Command: configure replace, Description: Restores the device configuration to its state prior to any changes made during the resolution process, using a previously saved configuration file.

- Step: Re-enable Power Supply Redundancy, Command: power supply redundancy, Description: Re-enables power supply redundancy if it was disabled during the replacement of a faulty power supply, ensuring the device returns to a fully redundant state.